Assignment 5 - Weather Data Analysis and Trend Identification using Python

Title:

Data Visualization for Weather Data Analysis and Trend Identification using Matplotlib and Seaborn

Aim:

To analyze and visualize weather data using Python libraries such as Matplotlib and Seaborn for better understanding of climate trends, seasonal patterns, and relationships between weather parameters.

Objectives:

1)To learn data visualization techniques in Python.

2)To represent weather data graphically.

3)To identify trends and seasonal patterns in weather data.

4)To create different types of statistical and categorical plots.

Algorithm:

1)Import required libraries such as Pandas, Matplotlib, and Seaborn.

2)Create a dataset containing monthly weather parameters.

3)Convert the dataset into a Pandas DataFrame.

4)Calculate average weather values for trend analysis.

5)Generate different types of visualizations:

- Line Plot
- Scatter Plot
- Histogram
- Density Plot
- Bar Chart
- Pie Chart
- Subplots
- Count Plot
- Heatmap
- Pair Plot
- 3D Plot

6)Add titles, labels, legends, and grid styles.

7)Display all graphs using plt.show().

8)Analyze the results obtained from the visualizations. Required graphs for weather data analysis and trend identification.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D
# Dataset
data = {
"Month" : ["Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"],
"Temperature": [22, 24, 28, 32, 36, 38, 35, 34, 31, 28, 25, 22],
"Humidity" : [80, 75, 65, 55, 45, 50, 70, 75, 72, 68, 78, 82],
"Rainfall" : [120, 90, 60, 30, 10, 5, 80, 100, 110, 95, 130, 140],
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"Wind_Speed" : [12, 14, 16, 18, 20, 22, 19, 17, 15, 13, 11, 10],
"AQI" : [85, 80, 75, 70, 65, 60, 72, 78, 82, 88, 90, 92],
"Season" : ["Winter","Winter","Spring","Spring","Summer","Summer",
"Monsoon","Monsoon","Monsoon","Autumn","Autumn","Winter"],
}
df = pd.DataFrame(data)
# Average Weather Value
df["Average"] = (df["Temperature"] + df["Humidity"] + df["Wind_Speed"]) / 3
sns.set_style("whitegrid")
# 1. Line Plot
plt.figure(figsize=(8,5))
plt.plot(df["Month"], df["Temperature"], marker='o', label="Temperature (°C)",
color='tomato')
plt.title("Monthly Temperature Trend")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.grid(True)
plt.show()
# 2. Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(df["Temperature"], df["Humidity"], color='steelblue', s=100,
edgecolors='black')
plt.title("Temperature vs Humidity")
plt.xlabel("Temperature (°C)")
plt.ylabel("Humidity (%)")
plt.show()
# 3. Histogram
plt.figure(figsize=(8,5))
plt.hist(df["Rainfall"], bins=6, color='steelblue', edgecolor='black')
plt.title("Rainfall Distribution")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Frequency")
plt.show()
# 4. Density Plot
plt.figure(figsize=(8,5))
sns.kdeplot(df["Temperature"], fill=True, color='tomato')
plt.title("Density Plot of Temperature")
plt.xlabel("Temperature (°C)")
plt.show()
# 5. Bar Chart
plt.figure(figsize=(8,5))
season_counts = df["Season"].value_counts()
plt.bar(season_counts.index, season_counts.values, color='orange',
edgecolor='black')
plt.title("Frequency of Seasons")
plt.xlabel("Season")
plt.ylabel("Count")
plt.show()
# 6. Pie Chart
season_counts = df["Season"].value_counts()
plt.figure(figsize=(6,6))
plt.pie(season_counts, labels=season_counts.index, autopct='%1.1f%%')
plt.title("Season Classification")
plt.show()
# 7. Subplots
fig, axes = plt.subplots(1, 2, figsize=(12,5))
axes[0].plot(df["Month"], df["Rainfall"], marker='o', color='steelblue')
axes[0].set_title("Monthly Rainfall Trend")
```

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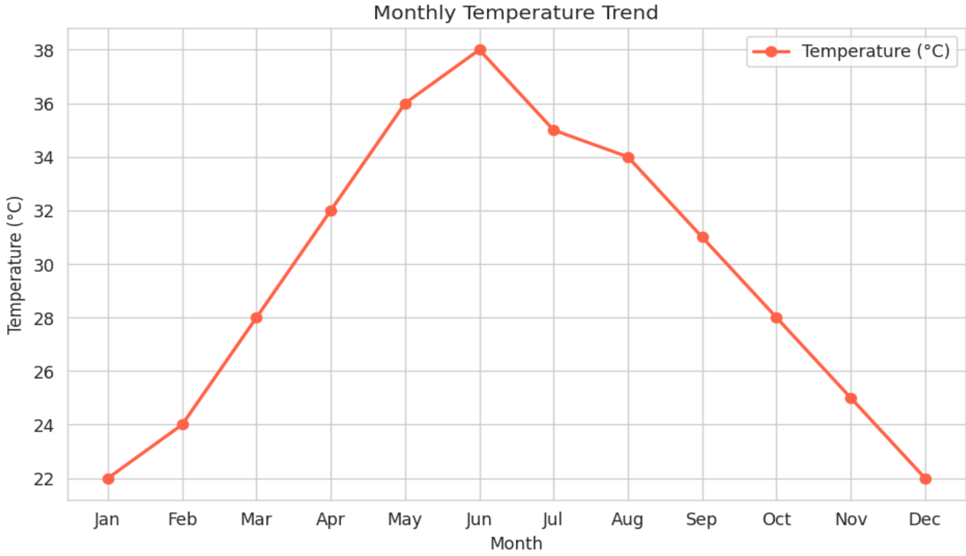
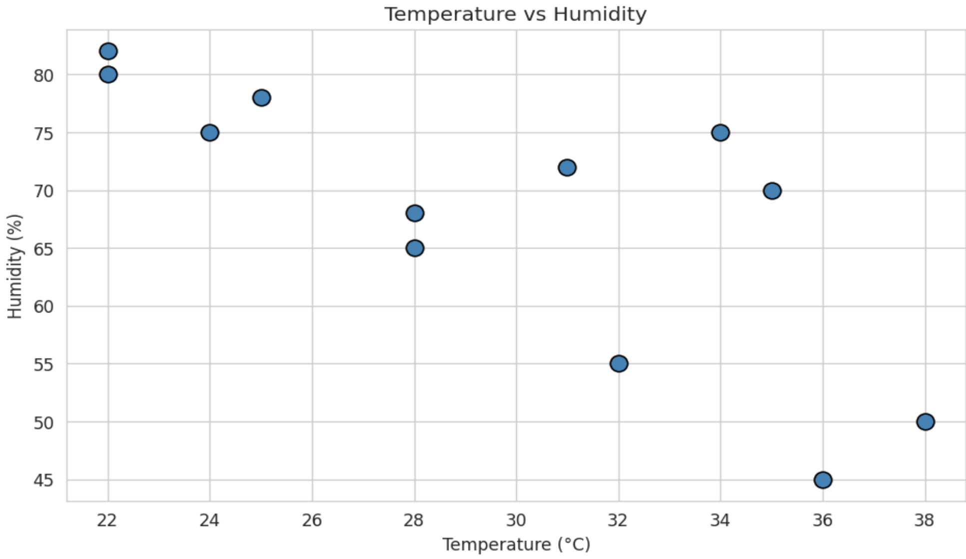
axes[1].bar(df["Month"], df["Wind_Speed"], color='mediumseagreen',
edgecolor='black')
axes[1].set_title("Monthly Wind Speed")
plt.tight_layout()
plt.show()
# 8. Count Plot
plt.figure(figsize=(8,5))
sns.countplot(x="Season", data=df)
plt.title("Count Plot of Seasons")
plt.show()
# 9. Heatmap
corr = df[["Temperature","Humidity","Rainfall","Wind_Speed","AQI"]].corr()
plt.figure(figsize=(7,5))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Weather Parameters Correlation Heatmap")
plt.show()
# 10. Pair Plot
sns.pairplot(df[["Temperature", "Humidity", "Rainfall", "Wind_Speed"]])
plt.show()
# 11. 3D Plot
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(df["Temperature"], df["Humidity"], df["Rainfall"])
ax.set_xlabel("Temperature (°C)")
ax.set_ylabel("Humidity (%)")
ax.set_zlabel("Rainfall (mm)")
plt.title("3D Weather Data Visualization")
plt.show()
```

Output

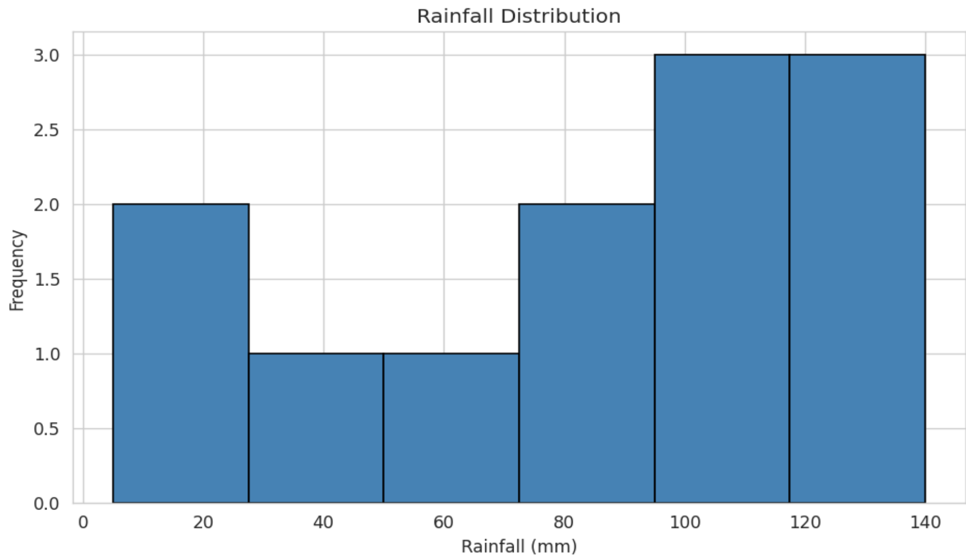
Line Plot



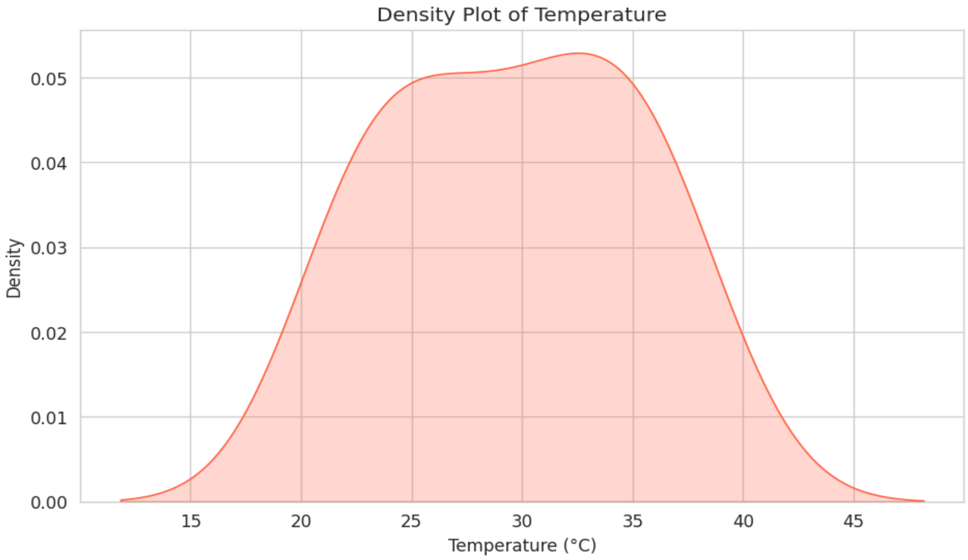
Scatter Plot



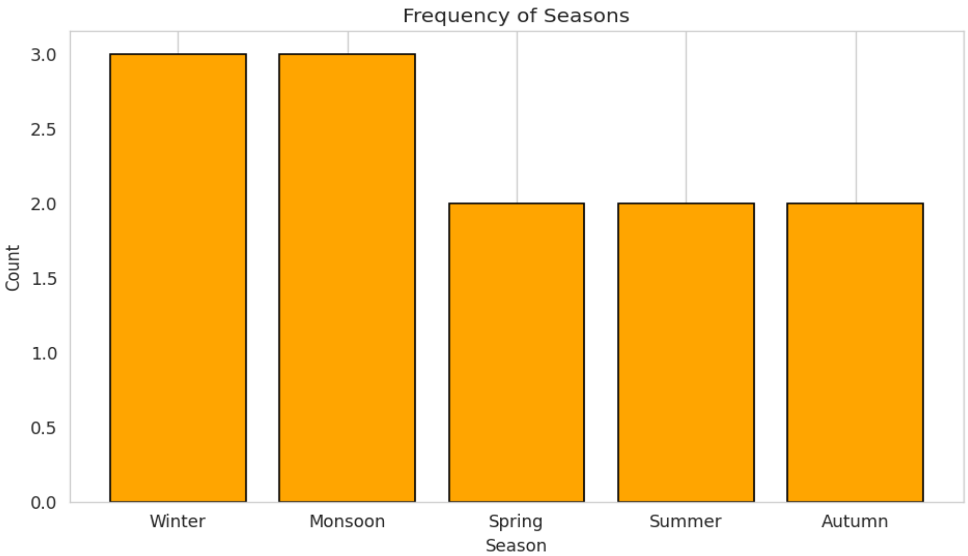
Histogram



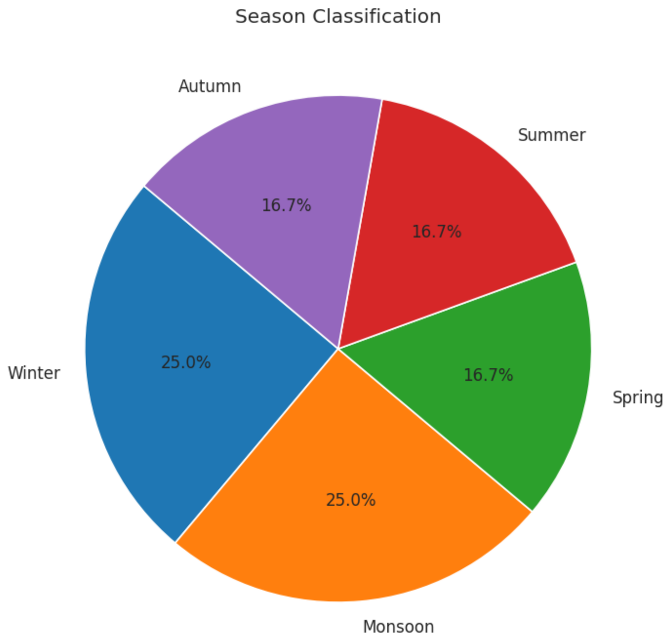
Density Plot



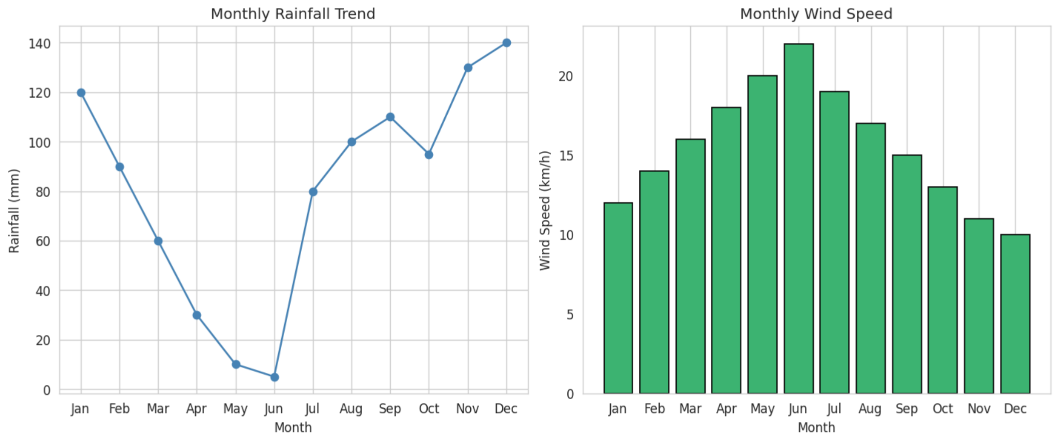
Bar Chart



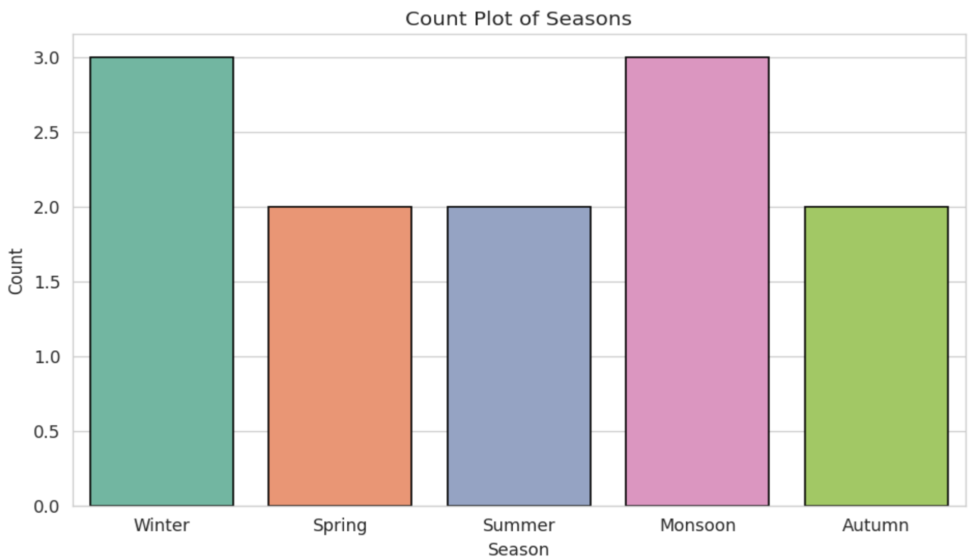
Pie Chart



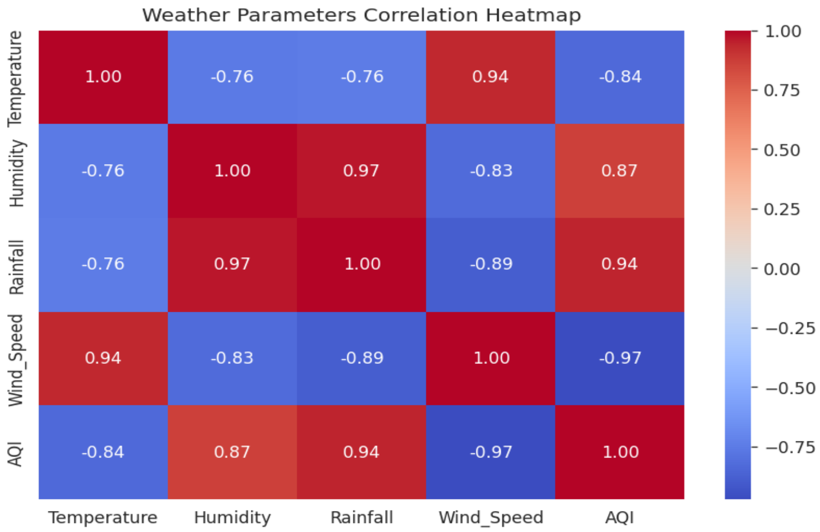
Subplots



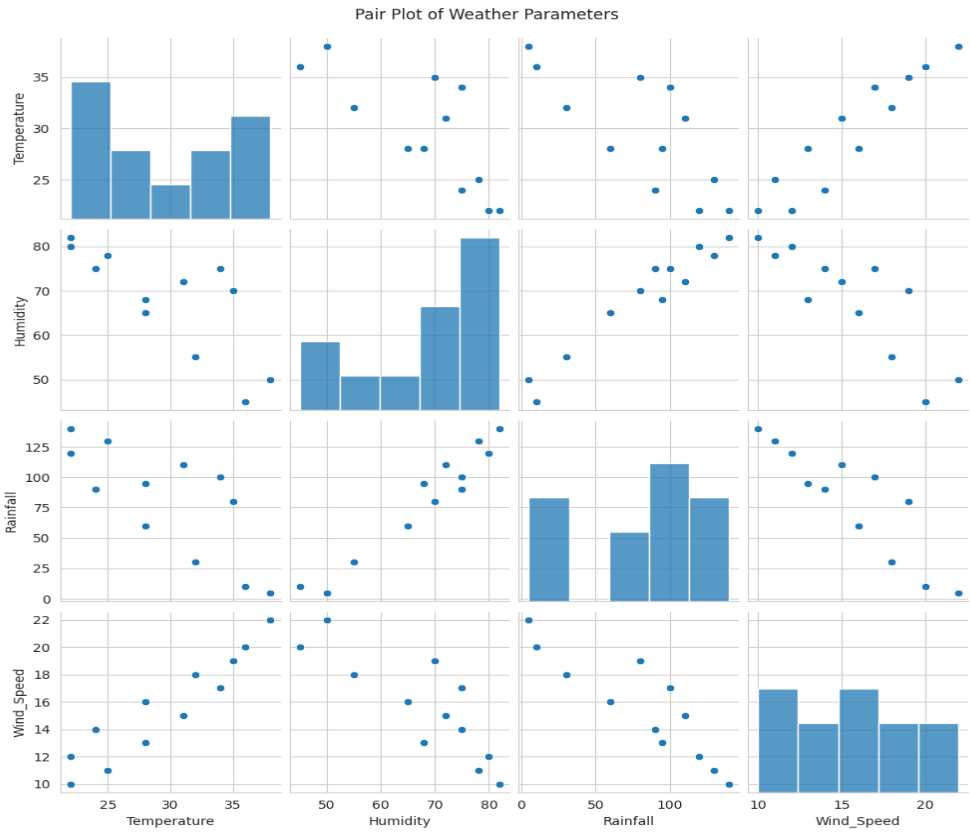
Count Plot



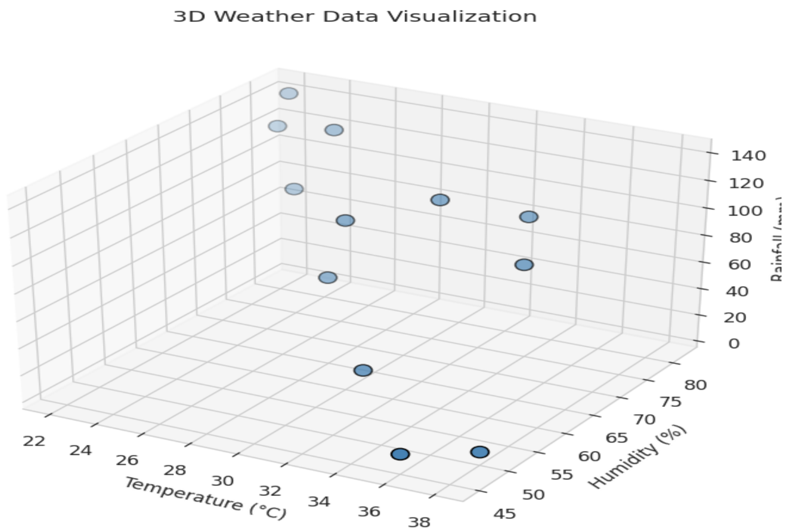
Heatmap


Weather Parameters Correlation Heatmap
Temperature
Humidity
Rainfall
Wind_Speed
AQI
1.00
-0.76
-0.76
0.94
-0.84
-0.76
1.00
0.97
-0.83
0.87
-0.76
0.97
1.00
-0.89
0.94
0.94
-0.83
-0.89
1.00
-0.97
-0.84
0.87
0.94
-0.97
1.00
Temperature
Humidity
Rainfall
Wind_Speed
AQI
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

Pair Plot



3D Plot



Result:

Thus, the weather data analysis and trend identification was successfully performed using Matplotlib and Seaborn. Different plots helped in understanding temperature trends, rainfall distributions, seasonal patterns, and relationships between weather parameters effectively.